

# European CRM Architecture Review.



Working architecture hypothesis  
for architecture review

*Fragmented client intelligence, missing cross-sell opportunities, and manual reporting.*

## Client intelligence is locked locally

### 1 – Fragmented data

Client knowledge sits with individual brokers in email, Excel trackers, and shared folders. If a broker in Madrid is tracking an investor whose strategy matches a Paris asset, neither side knows. That connection only emerges by chance, if it emerges at all.

## Reporting doesn't scale

### 2 – Manual process

Transaction reports are built manually in Excel. Price, buyer, seller, asset, and fee data assembled manually after every close. As deal volume grows, reporting burden grows with it.

# The Core Backbone | MVP Scope

*The MVP delivers a shared client intelligence foundation that every service line and every country inherits. Capital Markets is the first vertical deployed on it.*

## Shared Foundation

Designed to be reusable across service lines and countries

Accounts; Contacts; Properties; Engagement Tracking; Security

 Single Client Record

 Cross-SL Contact Intelligence

 Shared Property Records

 Outlook Integration

 SharePoint Auto-Provision

 Power BI Dashboards

 RBAC / Business Units

 Engagement History

 *Answers: "Who is this client to Knight Frank across every service line, every country?"*

## First Service-Line Build

### Capital Markets: First vertical on the shared foundation

- Capital Markets is the first vertical deployed on the shared foundation
- OSS, Valuations, etc. follow as separate Service Line solutions on the same backbone

### Key Principle: CRM Lite

*"The MVP is a shared client, contact, property and engagement foundation: a relationship and intelligence layer; not a full operational workflow system – which belongs to the Service Line deep build phase. Capital Markets is the first deeper workflow built on top."*

Proposed solution packages and ownership boundaries. Import order, dependencies and ownership need architecture validation.

|    |                                                                                                                                                                               |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L7 | <b>Cross-Border Overlays</b><br>KF_EIT; KF_Private_Office                                                                                                                     |
| L6 | <b>Service Line × Country</b><br>KF_CM_France; KF_CM_Spain; KF_OSS_France; ...                                                                                                |
| L5 | <b>Service Line Core</b><br>KF_CapitalMarkets_Core (8-stage Business Process Flow (BPF)); KF_OSS_Core; KF_Valuations_Core (future); KF_Marketing_Core; KF_Finance_Integration |
| L4 | <b>Country</b><br>KF_France; KF_Germany; KF_Poland; KF_Spain                                                                                                                  |
| L3 | <b>Region</b><br>KF_Europe – EUR default, GDPR baseline, Regional Power BI Dashboard                                                                                          |
| L2 | <b>KF_Core</b><br>40+ table shells; option sets; BU hierarchy; base security; Outlook sync; SharePoint; BI                                                                    |
| L1 | <b>Dataverse Platform (Microsoft-managed)</b><br>System tables; security framework; audit log                                                                                 |
| L0 | <b>External Systems</b><br>D365 CI-Journeys (same env); Finance (future)                                                                                                      |

Proposed Technical Rules

- > **KF\_Core Layer** defines ALL Option sets & ALL custom tables as base – schema name, primary key, and ownership model
- > Columns and relationships added by higher layers
- > No regional fork can **create a duplicate table**
- > Each layer is a separate **managed solution**
- > **No unmanaged customisations** in TEST, UAT, or PROD

Layer 6 Extension Rules

- Inherits the Service Line Core Layer BPF
  - Adds country-specific columns with prefixes: **kf\_fr\_** (France); **kf\_es\_** (Spain)
  - Adds regulatory stage gates that **block BPF progression**
  - Provides country-specific form layouts and option set visibility rules
  - NEVER renames or removes standard stages – only **ADDs** within them
- > Adding a country = **new L4 + L6 solutions** | Adding a service line = **new L5 solution**

## SECTION I

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# Scope & High-Level Requirements

What the MVP covers, and what it must deliver

# Implementation Scope and Future Expansion

What the MVP answers now, and what the future phases add later.

| DIMENSION         | MVP                                                                                         | Future Phases                                         |
|-------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Question Answered | "Who is this client?", "What does our EU pipeline look like?" and "How do we run CM deals?" | "How do we run every service line, in every country?" |
| Service Lines     | All SL Foundation; Capital Markets Deep Build                                               | + OSS, Valuations, Residential, UHNW                  |
| Markets           | France, Germany, Poland, Spain                                                              | Whole Europe                                          |
| Finance           | Manual (Finance Admin)                                                                      | Centralised EU ERP                                    |
| AI                | Copilot for Sale                                                                            | AI Builder + AI-powered matching                      |
| Marketing         | -                                                                                           | Full Customer Insights & Journeys                     |



## Core Master Data

- A single trusted view of property and clients across service lines and countries
- International standards for property and client type definitions



## Pipeline & Relationships

- Engagement tracking and cross-border unified client view
- Full pipeline visibility with built-in WIP data



## Productivity Tools

- Outlook sync
- SharePoint libraries
- Power BI dashboards
- Copilot for Sales



## Data Architecture

Layered Dataverse structure  
Layer 2 core, Layer 3 EUR defaults, Layer 4 country extensions, Layer 5 Service Line BPFs



## Security & Compliance

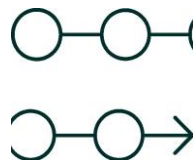
Cross-border data sharing agreement, EU data centres, GDPR consent baseline  
Role-based security, 7-year audit retention

# Capital Markets Requirements



## Deal Lifecycle & Entities

Standardized 8-stage BPF across all markets  
Custom entities for pitch tracking and transaction reports



## Reporting & Finance

Live pipeline visibility by stage, auto-populated reports  
WIP aging alerts and monthly reconciliation



## Investor Intelligence

Investor profiles with rule-based matching  
Cross-border visibility as a permanent feature



## Regulatory Compliance

Country-specific regulatory gates  
EIT cross-border KYC and billing



## SECTION II – Core Data Model

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# Client

# One Client, One View

Who we hold the relationship with, and who we contract with?

## ACCOUNT

OOB extended | Layer 2 Core

Organisation, brand, legal entity, fund, SPV or investor platform. Shared across every service line.

## CONTACT

OOB extended | Layer 2 Core

The individual person, always associated to an Account.

## ACCOUNT: CORE FIELDS

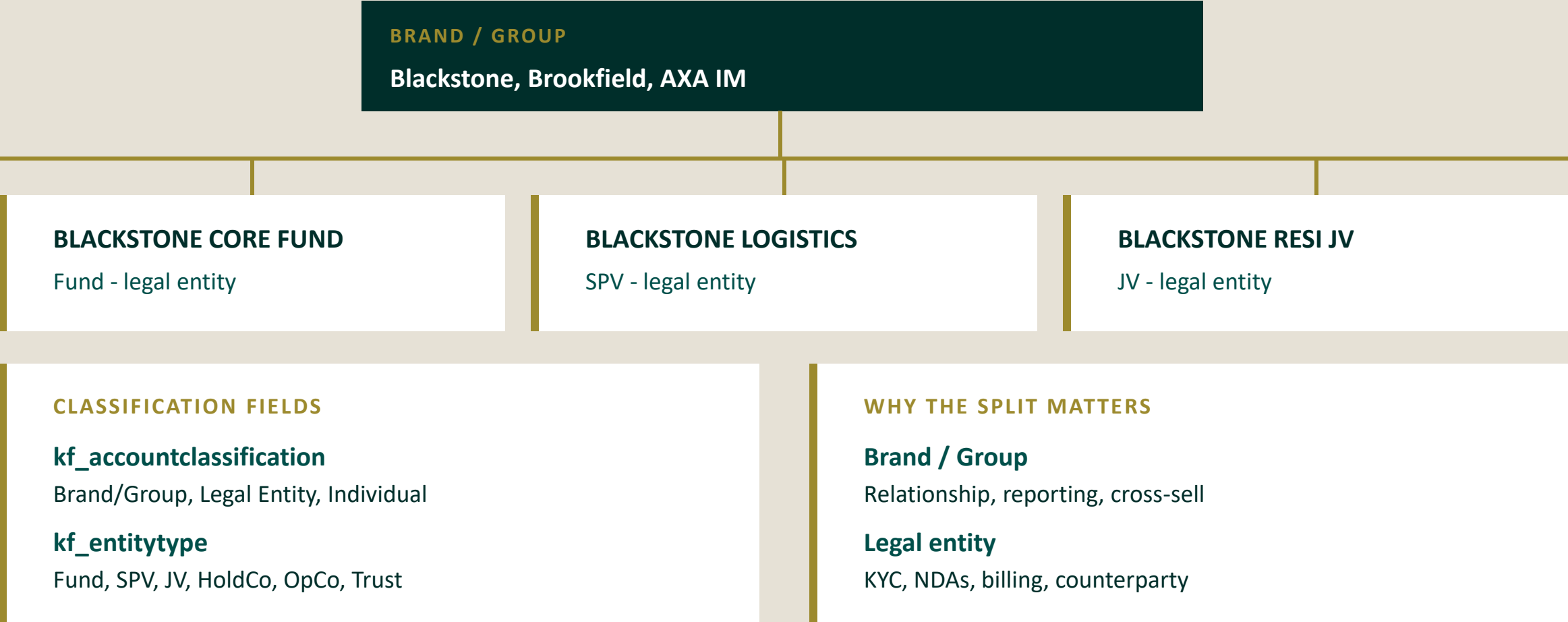
|                                   |                             |
|-----------------------------------|-----------------------------|
| <b>accountid</b>                  | GUID, primary key           |
| <b>name</b>                       | Organisation name           |
| <b>accounttype</b>                | Investor, vendor, occupier  |
| <b>address1_*</b>                 | Composite address           |
| <b>telephone1 / emailaddress1</b> | Primary contact details     |
| <b>owningbusinessunit</b>         | Security ownership          |
| <b>parentaccountid</b>            | Parent account hierarchy    |
| <b>kf_clientsector</b>            | SIC industry classification |
| <b>kf_companyreg</b>              | National register number    |

## CONTACT: CORE FIELDS

|                             |                          |
|-----------------------------|--------------------------|
| <b>contactid</b>            | GUID, primary key        |
| <b>firstname / lastname</b> | Person name              |
| <b>jobtitle</b>             | Role at the organisation |
| <b>emailaddress1</b>        | Primary email            |
| <b>telephone1</b>           | Primary phone            |
| <b>parentcustomerid</b>     | Lookup to Account        |
| <b>kf_processingconsent</b> | GDPR basis, Layer 3 EU   |

# Brand / Group vs. Legal Entity

A two-tier hierarchy gives one client view across many funds, SPVs and joint ventures.



# International Standards on Client Records

Two fields anchor every account to recognised classifications rather than free text.

## KF\_INDUSTRY

Industry classification mapped to SIC codes.

## KF\_REGISTRATIONNUMBER

Legal registration number from the national company register.

## THE STANDARDS

|               |                                         |
|---------------|-----------------------------------------|
| <b>SIC</b>    | UK and EU statistical industry codes    |
| <b>Levels</b> | Section, division and class codes       |
| <b>Use</b>    | Sector reporting, cross-sell, targeting |

## NATIONAL REGISTERS

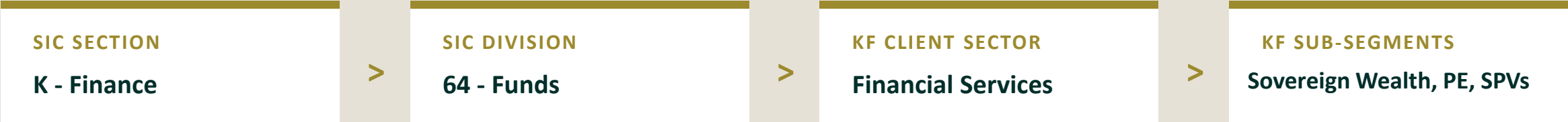
|                |                     |
|----------------|---------------------|
| <b>France</b>  | SIREN / SIRET       |
| <b>Germany</b> | Handelsregister HRB |
| <b>Spain</b>   | Registro Mercantil  |
| <b>Poland</b>  | KRS number          |

## WHY IT MATTERS

Standard codes make sector reporting comparable across Europe, and a validated registration number underpins KYC and invoicing.

# Client Industry Sectors: SIC to KF Mapping

SIC gives the common backbone; KF sectors describe how we actually serve clients.



See Appendix for the full mapping of 21 SIC sections to 66 KF client sectors and sub-segments.

## SECTION II – Core Data Model

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# Property

# Site, Property, DealProperty

Physical, commercial, residential and transactional facts are held on separate records.

## THE PHYSICAL ASSET

### kf\_Site

The canonical building or land parcel that every other record hangs off.

- kf\_uprn - UK reference
- kf\_cadastralref - EU registry
- kf\_landregistrytitle - deed
- kf\_loqateid - verified address

## THE COMMERCIAL RECORD

### kf\_Property

The shared asset register that every commercial service line reads.

- Identity and address, geocoded
- Sector, type, status, tenure
- Areas, floors, parking, planning
- Asking price, owner, ESG

## THE RESIDENTIAL RECORD

### kf\_ResProperty

The residential sibling on the same site, with its own saleable attributes.

- kf\_propertytype, kf\_tenure
- Bedrooms, bathrooms, receptions
- Weekly rent, guide price
- kf\_epcrating, kf\_siteid

## THE JUNCTION

### kf\_DealProperty

Deal-specific measures for one asset inside one deal, single or portfolio.

- kf\_allocationpercent, kf\_lotstatus
- kf\_passingrent, kf\_erv
- kf\_occupancy, kf\_wault
- kf\_capitalvalue\_psm, kf\_tenantid

## DESIGN PRINCIPLE

Stable characteristics live on the property records; deal-specific measures live on the junction, so a deal can never overwrite the master asset.

# Why Two Property Tables?

**kf\_Property**
**COMMERCIAL**

**KEY SCHEMA FIELDS**

Use class

Floor area

Lease terms

Occupational data

Investor profiles

**SERVICE LINES SERVED**

Capital Markets - deals, bids, DD, investors

OSS - engagements, leases, assessments

**DEAL CYCLE**

6-18 months for institutional buyers

3-12 months for corporate occupiers

**kf\_ResProperty**
**RESIDENTIAL**

**KEY SCHEMA FIELDS**

Bedrooms

Bathrooms

Tenure

Furnishing

Council tax

**SERVICE LINES SERVED**

Resi Sales, Lettings and Buying Agency

Resi Property Management

**DEAL CYCLE**

Days to weeks for individual buyers and tenants

Merging would create mostly blank columns for two unrelated audiences.

What they share is the physical building - which is what kf\_Site covers.

# Two Real Scenarios

Where one building is transacted by both the commercial and residential teams at once.



## SCENARIO 1

### Change of Use

An office block converts to residential. The commercial record is orphaned, a new resi record is created, and years of deal history are lost.

**Lost history: years of deal record left behind**



## SCENARIO 2

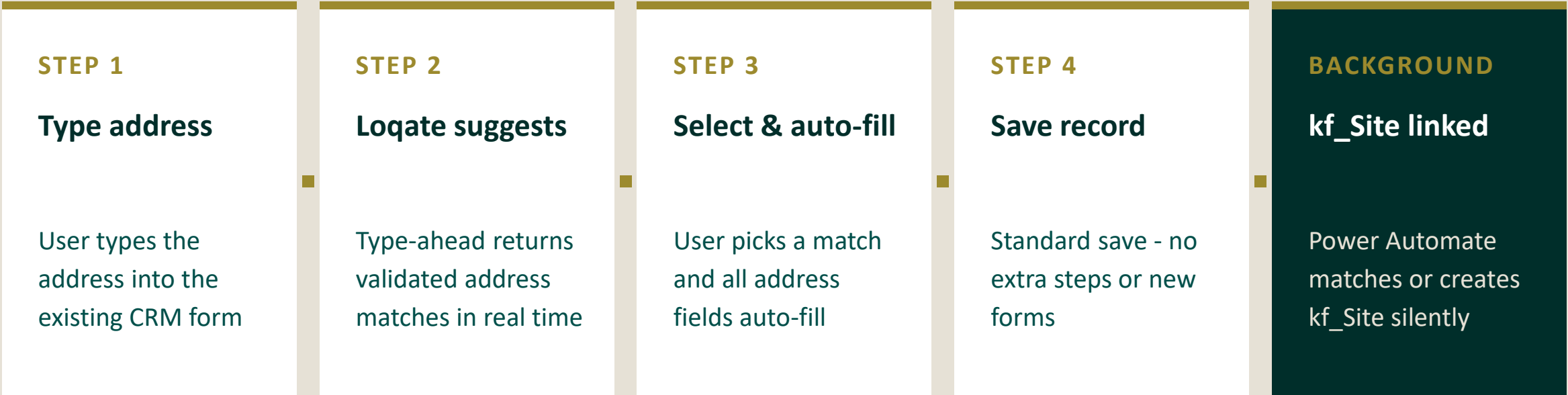
### Living Sectors — BTR

A Build-to-Rent block is sold as a CM asset, then let unit-by-unit. Duplicate records in both datasets keep cross-sell invisible.

**Data quality: duplicates and missed cross-sell**

# Auto Address Look Up with Loqate

Users do not manually create records for kf\_Site - matching and creation happen silently in the background.



# Every Service Line Reads the Same Record

One property record answers ownership, deals, valuations, leases and ESG in a single graph.

## HOW EACH SERVICE LINE CONNECTS

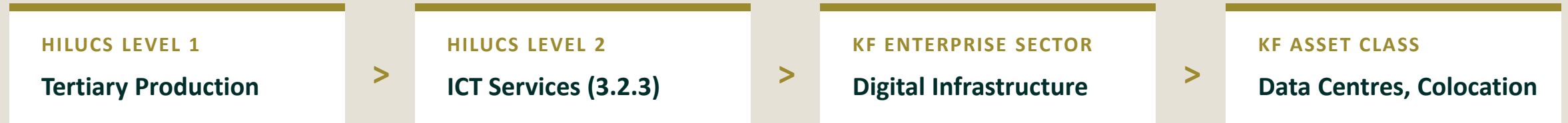
|                      |                            |              |
|----------------------|----------------------------|--------------|
| Capital Markets      | kf_DealProperty            | Many to many |
| OSS                  | kf_Engagement              | Many to one  |
| Valuations           | kf_ValuationInstruction    | Many to one  |
| ESG                  | kf_ESGAssessment           | Many to one  |
| Building Consultancy | kf_BuildingSurvey          | Many to one  |
| Development          | kf_DevelopmentProject      | Many to one  |
| Capital Advisory     | kf_DebtMandate             | Many to one  |
| Property Management  | kf_PropertyMandate         | Many to one  |
| Leasing              | kf_Lease                   | Many to one  |
| Residential          | kf_ResProperty via kf_Site | Sibling      |

## FROM ONE PROPERTY RECORD

- Who owns it?
- What deals has it been in?
- Who bid on it?
- What is its energy rating?
- Has it been valued?
- Is there a live lease?
- Is a development running?

# KF Asset Classes: INSPIRE HILUCS European Land-Use Standards

HILUCS gives the common backbone for Knight Frank asset classes.



See Appendix for the full mapping of all Knight Frank sectors and asset classes.

## SECTION II – Core Data Model

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# WIP

# WIP Functionality & User Journey

Bridging the gap between work delivered and revenue recognised



# WIP Data Model: CRM to Finance

Every CRM deal becomes a project in D365 Finance, and that bridge enables WIP reconciliation.

| CRM CONCEPT                                                                                                   | FINANCE CONCEPT                                                                                                         | WHY THEY MAP                                                                                                     |                                                                                                   |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| kf_WIP line                                                                                                   | Project                                                                                                                 | Both track the same billable engagement                                                                          |                                                                                                   |
| kf_FeeSchedule                                                                                                | Project budget / contract line                                                                                          | Both define expected revenue                                                                                     |                                                                                                   |
| BPF stage                                                                                                     | % complete                                                                                                              | Both measure progress to completion                                                                              |                                                                                                   |
| Deal status = Won                                                                                             | Milestone / invoice request                                                                                             | Both trigger billing                                                                                             |                                                                                                   |
| Account (client)                                                                                              | Customer (debtor)                                                                                                       | Both identify who pays                                                                                           |                                                                                                   |
| KF_WIP                                                                                                        |                                                                                                                         |                                                                                                                  |                                                                                                   |
| <ul style="list-style-type: none"><li>■ kf_wipid</li><li>■ kf_parenttype</li><li>■ kf_instructionid</li></ul> | <ul style="list-style-type: none"><li>■ kf_netfeetogroup</li><li>■ kf_officeretained</li><li>■ kf_probability</li></ul> | <ul style="list-style-type: none"><li>■ kf_grossfee</li><li>■ kf_reportingmonth</li><li>■ kf_wipstatus</li></ul> | <ul style="list-style-type: none"><li>■ kf_invoicenumbr</li><li>■ kf_fin_localsystemref</li></ul> |

# The 8-Stage Process Drives WIP

Each deal stage maps to a finance percentage complete that automates the WIP calculation.

|    |     |    |     |    |     |    |      |
|----|-----|----|-----|----|-----|----|------|
| S1 | 5%  | S2 | 15% | S3 | 25% | S4 | 35%  |
| S5 | 50% | S6 | 65% | S7 | 85% | S8 | 100% |

## HOW THE INTEGRATION FLOWS RUN

- S3 fires Flow 1: the Finance project is created
- Deal won raises a draft invoice request
- A monthly batch refreshes WIP balance and ageing
- Each stage change updates % complete in Finance
- Invoice and payment data returns within hours
- A credit hold alerts the broker in Teams

## SECTION III

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# Capital Markets Data Model

How deals, bids and approvals are captured across the eight-stage lifecycle

# Capital Markets Data Model

Twelve entities capture every stage of the deal lifecycle, from pitch to completion



## Origination & Agreement

- **kf\_Deal**  
Core deal record with full lifecycle tracking
- **kf\_Pitch**  
Competitive pitch tracking
- **kf\_NDA**  
NDA management



## Bidding & Due Diligence

- **kf\_Bid**  
Multi-round bid management
- **kf\_DDMilestone**  
Due diligence milestone tracking
- **kf\_RedFlag**  
Risk and red flag tracking



## Compliance & Investors

- **kf\_KYCRecord**  
KYC/AML compliance tracking
- **kf\_InvestorProfile**  
Investor preferences and strategy
- **kf\_DataRoomAccess**  
Data room access management

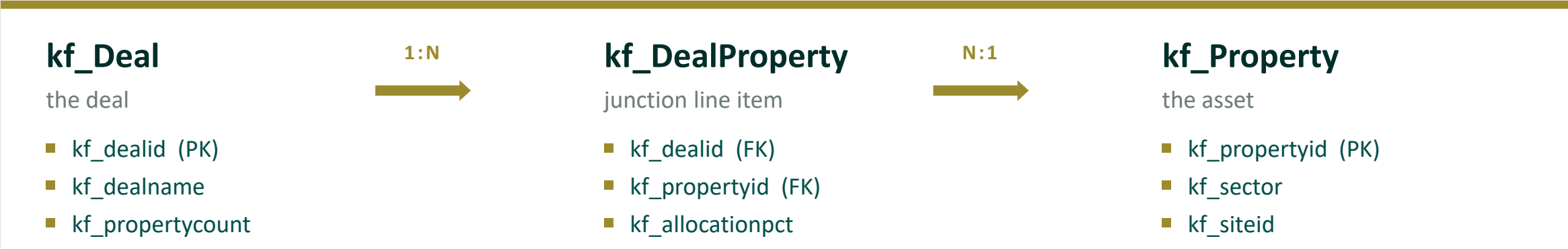


## Commercials & Reporting

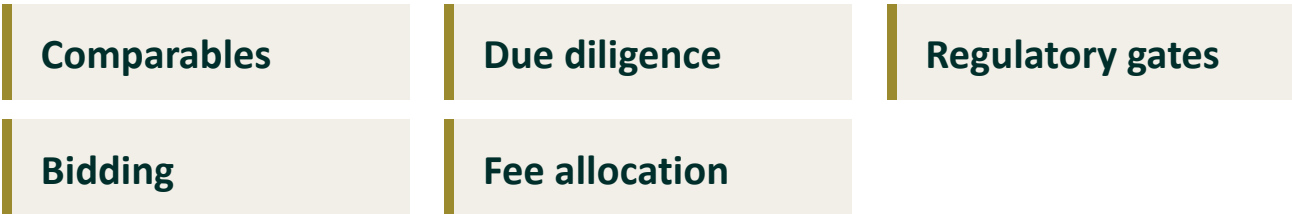
- **kf\_FeeSchedule**  
Fee structure management
- **kf\_TransactionReport**  
Auto-generated transaction reports
- **kf\_DealProperty**  
Portfolio property-level tracking

# Portfolio Data Model & Workflow

A junction record links one deal to many assets, so a portfolio stays a single transaction.



## TRACKED PER PROPERTY



## WHY IT STAYS SIMPLE FOR BROKERS

- Sub-grid only on multi-asset deals
- DD status rolls up to deal level
- Gates fire automatically per market
- Portfolios close as one transaction

# Client Touchpoints Across the Lifecycle

Each entity points at the right level: the group for relationships, the legal entity for compliance & billing.

## DEAL

**kf\_accountid**

Brand/Group client organisation

**kf\_contactid**

Primary deal contact

**kf\_legalentityaccountid**

Contractual counterparty, required at mandate

## COMPLIANCE

**kf\_NDA**

Counterparty legal entity plus signatory

**kf\_KYCRecord**

AML checks at legal entity level

**Brand/Group**

Explicitly excluded from NDA records

## BID

**kf\_investoraccountid**

The fund, SPV or JV bidding

**kf\_investorcontactid**

The individual contact

**kf\_investorprofileid**

Investment profile behind the bid

## CONTACT ROLES

Contacts play contextual roles through entity lookups: deal contact, NDA signatory, vendor, buyer, landlord, tenant.

## GDPR LAYER, EUROPE

Contacts carry consent, subscription and segmentation data. kf\_processingconsent covers legitimate interest, consent, legal obligation or not assessed.

# kf\_InvestorProfile: Knowing What Clients Want

Investment strategy sits on the Brand/Group account, so one group can hold several profiles.



## SAMPLE PROFILE: BLACKSTONE VALUE ADD

|                        |                                                              |
|------------------------|--------------------------------------------------------------|
| <b>Strategy</b>        | Core / Core-Plus / Value-Add / Opportunistic                 |
| <b>Target sectors</b>  | Offices, Logistics, Residential, Data Centres                |
| <b>Geographies</b>     | UK, France, Germany, Nordics, Iberia, CEE                    |
| <b>Ticket size</b>     | kf_ticketsize <sub>min</sub> to kf_ticketsize <sub>max</sub> |
| <b>Yield &amp; ESG</b> | Target yield, ESG mandatory flag                             |
| <b>Status</b>          | Active buyer / Watching / Paused / Inactive                  |

## RELATIONSHIP TRACKING

Owner and last contact date  
Rollups: deals and total invested

## CLIENT STRATEGY

Becomes the formal investor strategy repository for the region.

# Capital Markets Regulatory Gates

Local approval gates sit inside the standard eight-stage deal lifecycle.

## EIGHT-STAGE DEAL LIFECYCLE

|    |                 |                                                                                                                                                        |                                  |
|----|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| S1 | Origination     |                                                                                                                                                        |                                  |
| S2 | Pitch & Mandate | <ul style="list-style-type: none"> <li>Conflict check</li> </ul>                                                                                       | EIT                              |
| S3 | Instruction     | <ul style="list-style-type: none"> <li>Multi-jurisdiction KYC</li> </ul>                                                                               | EIT                              |
| S4 | Marketing       |                                                                                                                                                        |                                  |
| S5 | Bidding         |                                                                                                                                                        |                                  |
| S6 | Exclusivity     |                                                                                                                                                        |                                  |
| S7 | Due Diligence   | <ul style="list-style-type: none"> <li>City pre-emption</li> <li>Notarial process</li> <li>Right of refusal</li> <li>Pre-emption delegation</li> </ul> | France<br>France<br>Spain<br>EIT |
| S8 | Completion      | <ul style="list-style-type: none"> <li>Notarial deed signing</li> <li>Intercompany billing</li> </ul>                                                  | France<br>EIT                    |

## COUNTRY-SPECIFIC GATES

### Local rules, standard lifecycle



- France: city pre-emption and notarial deed
- Spain: right-of-refusal on listed buildings

## EIT

### The cross-border team



- Multi-jurisdiction KYC and conflict checks
- Intercompany billing across markets

Current hypothesis: Power Apps model-driven apps on Dataverse are a better fit where the value sits in a custom Knight Frank data model rather than a packaged sales process.

✖ Where D365 Sales may be a poor fit

Packaged CRM assumes a lifecycle that differs from KF advisory

- D365 model: Lead → Opportunity → Quote → Order → Invoice
- KF lifecycle (Capital Markets): Competitive pitching; NDA issuance; Multi-round bidding; Due diligence with milestone tracking; Regulatory gates per jurisdiction

D365 Sales entity fit – 11 core KF entities

| Entity                                                                                                    | D365 OOB fit      |
|-----------------------------------------------------------------------------------------------------------|-------------------|
| Opportunity, Quote, Product/Price List                                                                    | Poor fit (3)      |
| kf_Pitch, kf_NDA, kf_Bid, kf_DDMilestone, kf_RedFlag, kf_InvestorProfile, kf_DataRoomAccess, kf_KYCRecord | No equivalent (8) |

Unwanted features bundled in D365 Sales

- Product Catalogue;
- LinkedIn Sales Navigator

Current view

Power Apps model-driven apps appear to cover the core platform capabilities for a custom CRM foundation, subject to validation on licensing, restricted tables and Copilot.

✔ What Dataverse Provides Natively

Core platform capabilities (to validate)

- ✔ Custom tables with relationships
- ✔ Business Process Flows (BPFs)
- ✔ Security roles + Business Units
- ✔ Power Automate
- ✔ Power BI
- ✔ Outlook / Teams integration
- ✔ SharePoint document management
- ✔ Dataverse audit trail
- ✔ API access (OData + custom)
- ✔ Mobile access

# Appendix: Client Industry Taxonomy - SIC to KF Mapping (1 of 2)

| SEC | SIC SECTION                    | DIVISIONS              | KF CLIENT SECTOR                                  | KF SUB-SEGMENTS                                                                |
|-----|--------------------------------|------------------------|---------------------------------------------------|--------------------------------------------------------------------------------|
| A   | Agriculture, Forestry and Fish | 1                      | <b>Agriculture &amp; Farming</b>                  | Arable Farms, Livestock, Mixed Farming, Estate Owners, Agricultural Cooperativ |
| A   | Agriculture, Forestry and Fish | 2                      | <b>Forestry &amp; Woodland</b>                    | Timber Companies, Forest Management, Carbon Forestry Operators                 |
| A   | Agriculture, Forestry and Fish | 3                      | <b>Marine &amp; Aquaculture</b>                   | Fisheries Operators, Aquaculture Companies                                     |
| B   | Mining and Quarrying           | 5, 7, 8, 9             | <b>Mining &amp; Extraction</b>                    | Coal Mining Operators                                                          |
| B   | Mining and Quarrying           | 6                      | <b>Oil &amp; Gas</b>                              | Upstream Operators, E&P Companies                                              |
| C   | Manufacturing                  | 10, 11                 | <b>Food &amp; Beverage Manufacturing</b>          | Food Producers, Processors, Packagers                                          |
| C   | Manufacturing                  | 12, 13, 14, 15, 31, 32 | <b>Consumer Goods Manufacturing</b>               | Tobacco Companies                                                              |
| C   | Manufacturing                  | 16, 17, 18, 22, 23, 25 | <b>Industrial Manufacturing</b>                   | Timber Processing, Wood Products                                               |
| C   | Manufacturing                  | 19, 20                 | <b>Energy &amp; Chemicals</b>                     | Refineries, Petrochemical Plants                                               |
| C   | Manufacturing                  | 21                     | <b>Life Sciences &amp; Pharmaceuticals</b>        | Pharma Companies, Biotech, Drug Manufacturing                                  |
| C   | Manufacturing                  | 24                     | <b>Heavy Industry</b>                             | Steelworks, Smelters, Metal Production                                         |
| C   | Manufacturing                  | 26, 27                 | <b>Technology Manufacturing</b>                   | Semiconductor, Electronics, Optics                                             |
| C   | Manufacturing                  | 28                     | <b>Advanced Manufacturing</b>                     | Machine Builders, Capital Goods                                                |
| C   | Manufacturing                  | 29                     | <b>Automotive</b>                                 | Vehicle Manufacturers, OEMs, Tier 1 Suppliers                                  |
| C   | Manufacturing                  | 30                     | <b>Aerospace &amp; Defence</b>                    | Aircraft, Shipbuilding, Rail Manufacturing                                     |
| C   | Manufacturing                  | 33                     | <b>Industrial Services</b>                        | Maintenance Providers, Equipment Installers                                    |
| D   | Electricity, Gas, Steam and AC | 35                     | <b>Energy &amp; Utilities</b>                     | Power Companies, Gas Distributors, District Heating, Renewable Generators      |
| E   | Water Supply, Sewerage, Waste  | 36, 37, 38, 39         | <b>Environmental Services &amp; Utilities</b>     | Water Companies, Treatment Operators                                           |
| F   | Construction                   | 41                     | <b>Real Estate Development &amp; Construction</b> | House Builders, Commercial Developers, Design & Build                          |
| F   | Construction                   | 42                     | <b>Infrastructure &amp; Civil Engineering</b>     | Civil Engineers, Infrastructure Contractors                                    |
| F   | Construction                   | 43                     | <b>Construction Services</b>                      | Electrical Contractors, Plumbing, Fitout, Demolition                           |
| G   | Wholesale and Retail Trade     | 45                     | <b>Automotive Retail</b>                          | Motor Dealers, Vehicle Distribution, Showrooms                                 |
| G   | Wholesale and Retail Trade     | 46                     | <b>Wholesale &amp; Distribution</b>               | Wholesalers, Import/Export, B2B Distribution                                   |
| G   | Wholesale and Retail Trade     | 47                     | <b>Retail &amp; Consumer</b>                      | High Street Retailers, Supermarkets, E-commerce, Convenience, Discount         |
| H   | Transportation and Storage     | 49                     | <b>Transport &amp; Mobility</b>                   | Rail Operators, Bus Companies, Road Freight, Pipeline Operators                |
| H   | Transportation and Storage     | 50                     | <b>Shipping &amp; Maritime</b>                    | Shipping Lines, Ferry Operators, Inland Waterway                               |
| H   | Transportation and Storage     | 51                     | <b>Aviation</b>                                   | Airlines, Cargo Airlines, Charter Operators                                    |
| H   | Transportation and Storage     | 52                     | <b>Logistics &amp; Distribution</b>               | 3PL, Warehousing Operators, Freight Forwarders, Fulfilment                     |
| H   | Transportation and Storage     | 53                     | <b>Postal &amp; Courier</b>                       | National Mail, Parcel Companies, Last-Mile Delivery                            |
| I   | Accommodation and Food Service | 55                     | <b>Hotels &amp; Hospitality</b>                   | Hotel Operators, Resort Groups, Serviced Apartment Operators, Hostels          |
| I   | Accommodation and Food Service | 56                     | <b>Food &amp; Beverage Operators</b>              | Restaurant Groups, QSR Chains, Pub Companies, Caterers, Cafés                  |
| J   | Information and Communication  | 58                     | <b>Media &amp; Publishing</b>                     | Publishers, Digital Media, News Organisations                                  |
| J   | Information and Communication  | 59, 60                 | <b>Media &amp; Entertainment</b>                  | Film Studios, Production Companies, Streaming Platforms                        |

# Appendix: Client Industry Taxonomy - SIC to KF Mapping (2 of 2)

| SEC | SIC SECTION                    | DIVISIONS | KF CLIENT SECTOR                           | KF SUB-SEGMENTS                                                                |
|-----|--------------------------------|-----------|--------------------------------------------|--------------------------------------------------------------------------------|
| J   | Information and Communication  | 61        | <b>Telecommunications</b>                  | Mobile Operators, ISPs, Fibre Providers, Tower Companies                       |
| J   | Information and Communication  | 62, 63    | <b>Technology</b>                          | Software Companies, IT Consultancies, SaaS, PropTech, Cyber Security           |
| K   | Financial and Insurance Activi | 64        | <b>Financial Services &amp; Investment</b> | Banks, Building Societies, Investment Funds, Holding Companies, PE, VC, SWFs,  |
| K   | Financial and Insurance Activi | 65        | <b>Insurance &amp; Pensions</b>            | Insurers, Reinsurers, Pension Funds, Life Companies                            |
| K   | Financial and Insurance Activi | 66        | <b>Financial Services Support</b>          | Fund Administration, Brokerage, Financial Advisory, Wealth Management          |
| L   | Real Estate Activities         | 68        | <b>Real Estate</b>                         | REITs, Property Companies, Asset Managers, Developers, Agents, Landlords, Prop |
| M   | Professional, Scientific and T | 69        | <b>Legal &amp; Accounting</b>              | Law Firms, Accountancy Practices, Tax Advisory                                 |
| M   | Professional, Scientific and T | 70        | <b>Professional Services</b>               | Management Consultancies, Corporate HQs, Strategy Firms                        |
| M   | Professional, Scientific and T | 71        | <b>Design &amp; Engineering</b>            | Architects, Engineers, Surveyors, Testing Laboratories                         |
| M   | Professional, Scientific and T | 72        | <b>Research &amp; Development</b>          | R&D Centres, Research Institutes, Biotech R&D                                  |
| M   | Professional, Scientific and T | 73        | <b>Marketing &amp; Creative</b>            | Advertising Agencies, Market Research, PR Firms                                |
| M   | Professional, Scientific and T | 74        | <b>Specialist Professional Services</b>    | Design Studios, Photography, Translation, Consulting                           |
| M   | Professional, Scientific and T | 75        | <b>Veterinary</b>                          | Veterinary Practices, Animal Hospitals                                         |
| N   | Administrative and Support Ser | 77        | <b>Rental &amp; Leasing</b>                | Vehicle Leasing, Equipment Rental, Plant Hire                                  |
| N   | Administrative and Support Ser | 78        | <b>Staffing &amp; Recruitment</b>          | Recruitment Agencies, Temp Staffing, HR Outsourcing                            |
| N   | Administrative and Support Ser | 79        | <b>Travel &amp; Tourism</b>                | Travel Agencies, Tour Operators, OTAs                                          |
| N   | Administrative and Support Ser | 80        | <b>Security Services</b>                   | Security Companies, Investigation Services, Guarding                           |
| N   | Administrative and Support Ser | 81        | <b>Facilities Management</b>               | FM Companies, Cleaning, Landscaping, Building Maintenance                      |
| N   | Administrative and Support Ser | 82        | <b>Business Support Services</b>           | BPO, Call Centres, Shared Service Centres                                      |
| O   | Public Administration and Defe | 84        | <b>Government &amp; Public Sector</b>      | Central Government, Local Authorities, Defence, Emergency Services, Diplomatic |
| P   | Education                      | 85        | <b>Education</b>                           | Universities, Schools, Colleges, Training Providers, EdTech, Academy Trusts    |
| Q   | Human Health and Social Work   | 86        | <b>Healthcare</b>                          | NHS Trusts, Private Hospitals, GP Practices, Dental Chains, Diagnostics        |
| Q   | Human Health and Social Work   | 87        | <b>Care &amp; Senior Living</b>            | Care Home Operators, Nursing Home Groups, Assisted Living, Retirement Villages |
| Q   | Human Health and Social Work   | 88        | <b>Social Services</b>                     | Charities, Social Enterprises, Community Services                              |
| R   | Arts, Entertainment and Recrea | 90        | <b>Creative &amp; Cultural</b>             | Theatres, Music Venues, Art Galleries, Creative Studios                        |
| R   | Arts, Entertainment and Recrea | 91        | <b>Cultural Institutions</b>               | Museums, Libraries, Heritage Organisations                                     |
| R   | Arts, Entertainment and Recrea | 92        | <b>Gaming &amp; Betting</b>                | Casinos, Betting Shops, Online Gambling Operators                              |
| R   | Arts, Entertainment and Recrea | 93        | <b>Sports &amp; Leisure</b>                | Gym Operators, Sports Clubs, Leisure Centres, Golf Clubs                       |
| S   | Other Service Activities       | 94        | <b>Membership &amp; Associations</b>       | Trade Bodies, Professional Associations, Trade Unions, Religious Organisations |
| S   | Other Service Activities       | 95        | <b>Repair Services</b>                     | IT Repair, Household Appliance Repair                                          |
| S   | Other Service Activities       | 96        | <b>Personal Services</b>                   | Hairdressers, Spas, Funeral Services, Laundry                                  |
| T   | Activities of Households as Em | 97, 98    | <b>UHNW &amp; Private Individuals</b>      | UHNW Individuals, Family Offices, Private Estates                              |
| U   | Extraterritorial Organisations | 99        | <b>International &amp; Diplomatic</b>      | Embassies, International Organisations, UN Bodies, EU Institutions, NATO       |

# Appendix: Master Taxonomy - HILUCS to KF Mapping (1 of 2)

| L1 | HILUCS LEVEL 1       | L2    | HILUCS LEVEL 2                               | KF SECTOR                        | KF ASSET CLASSES                                                    |
|----|----------------------|-------|----------------------------------------------|----------------------------------|---------------------------------------------------------------------|
| 1  | Primary Production   | 1.1   | Agriculture                                  | <b>Agriculture</b>               | Arable, Livestock, Mixed Farming, Orchards, Vineyards +             |
| 1  | Primary Production   | 1.2   | Forestry                                     | <b>Forestry</b>                  | Timber Estates, Managed Woodland, Carbon Forestry, Biomass Forestry |
| 1  | Primary Production   | 1.3   | Mining & Quarrying                           | <b>Natural Resources</b>         | Quarry, Aggregate Extraction, Mining Operations                     |
| 1  | Primary Production   | 1.4   | Aquaculture & Fishing                        | <b>Marine Resources</b>          | Fisheries, Aquaculture                                              |
| 1  | Primary Production   | 1.5   | Other Primary Production                     | <b>Rural &amp; Land Holdings</b> | Sporting Estates, Hunting Estates, Managed Rural Land               |
| 2  | Secondary Production | 2.1   | Raw Industry                                 | <b>Industrial Manufacturing</b>  | Chemical Plants, Steelworks, Cement Works, Processing Plants        |
| 2  | Secondary Production | 2.2   | Heavy End Product Industry                   | <b>Advanced Manufacturing</b>    | Automotive, Aerospace, Engineering Manufacturing                    |
| 2  | Secondary Production | 2.3   | Light End Product Industry                   | <b>Light Industrial</b>          | Food Production, Consumer Manufacturing, Electronics Production     |
| 2  | Secondary Production | 2.4   | Energy Production                            | <b>Energy</b>                    | Power Stations, Solar Farms, Wind Farms, Battery Storage +          |
| 2  | Secondary Production | 2.5   | Other Industry                               | <b>Specialist Industrial</b>     | Defence Manufacturing, Specialist Production Facilities             |
| 3  | Tertiary Production  | 3.1.1 | Wholesale & Retail Trade                     | <b>Retail</b>                    | Shopping Centres, Retail Parks, High Street, Outlet Centres +       |
| 3  | Tertiary Production  | 3.1.2 | Real Estate Services                         | <b>Mixed Commercial</b>          | Mixed Use Commercial Developments                                   |
| 3  | Tertiary Production  | 3.1.3 | Accommodation & Food Services                | <b>Hotels &amp; Hospitality</b>  | Hotels, Resorts, Aparthotels, Serviced Apartments, Extended Stay    |
| 3  | Tertiary Production  | 3.1.3 | Accommodation & Food Services                | <b>Food &amp; Beverage</b>       | Restaurants, Bars, Cafés, Pubs                                      |
| 3  | Tertiary Production  | 3.2.2 | Professional Technical & Scientific Services | <b>Office</b>                    | Prime Office, Grade A Office, Flexible Workspace, Corporate HQ +    |
| 3  | Tertiary Production  | 3.2.2 | Professional Technical & Scientific Services | <b>Life Sciences</b>             | Research Campuses, Laboratory Space, Wet Labs, Science Parks        |
| 3  | Tertiary Production  | 3.2.3 | Information & Communication Services         | <b>Digital Infrastructure</b>    | Data Centres, Hyperscale, Colocation, Edge Facilities               |
| 3  | Tertiary Production  | 3.2.3 | Information & Communication Services         | <b>Telecom Infrastructure</b>    | Exchanges, Telecom Hubs, Fibre Facilities                           |
| 3  | Tertiary Production  | 3.2.1 | Financial & Insurance Services               | <b>Financial Services</b>        | Banking HQs, Financial District Offices                             |
| 3  | Tertiary Production  | 3.3.2 | Educational Services                         | <b>Education</b>                 | Schools, Universities, Colleges, Training Centres                   |
| 3  | Tertiary Production  | 3.3.3 | Health & Social Services                     | <b>Healthcare</b>                | Hospitals, Medical Centres, Clinics                                 |
| 3  | Tertiary Production  | 3.3.3 | Health & Social Services                     | <b>Care &amp; Senior Living</b>  | Care Homes, Nursing Homes, Assisted Living                          |
| 3  | Tertiary Production  | 3.3.1 | Public Administration Services               | <b>Public Sector</b>             | Government Buildings, Civic Estate, Local Authority Buildings       |
| 3  | Tertiary Production  | 3.4.1 | Cultural Services                            | <b>Culture &amp; Heritage</b>    | Museums, Galleries, Heritage Assets                                 |
| 3  | Tertiary Production  | 3.4.2 | Entertainment Services                       | <b>Leisure</b>                   | Cinemas, Casinos, Theme Parks                                       |
| 3  | Tertiary Production  | 3.4.3 | Sports Infrastructure                        | <b>Sports &amp; Stadia</b>       | Stadiums, Arenas, Golf Courses, Sports Complexes                    |
| 3  | Tertiary Production  | 3.4.4 | Open Air Recreation                          | <b>Recreation</b>                | Parks, Visitor Attractions, Recreation Grounds                      |

# Appendix: Master Taxonomy - HILUCS to KF Mapping (2 of 2)

| L1 | HILUCS LEVEL 1                   | L2    | HILUCS LEVEL 2                          | KF SECTOR                           | KF ASSET CLASSES                                            |
|----|----------------------------------|-------|-----------------------------------------|-------------------------------------|-------------------------------------------------------------|
| 4  | Transport, Logistics & Utilities | 4.1.1 | Road Transport                          | <b>Road Infrastructure</b>          | Service Stations, Major Road Assets                         |
| 4  | Transport, Logistics & Utilities | 4.1.2 | Railway Transport                       | <b>Rail Infrastructure</b>          | Stations, Rail Hubs, Freight Rail Assets                    |
| 4  | Transport, Logistics & Utilities | 4.1.3 | Air Transport                           | <b>Aviation</b>                     | Airports, Cargo Airports, Airfields                         |
| 4  | Transport, Logistics & Utilities | 4.1.4 | Water Transport                         | <b>Maritime</b>                     | Ports, Marinas, Container Terminals                         |
| 4  | Transport, Logistics & Utilities | 4.2   | Logistical & Storage Services           | <b>Logistics</b>                    | Big Box Logistics, Distribution Centres, Fulfilment Centres |
| 4  | Transport, Logistics & Utilities | 4.2   | Logistical & Storage Services           | <b>Urban Logistics</b>              | Last Mile Logistics, Parcel Hubs                            |
| 4  | Transport, Logistics & Utilities | 4.2   | Logistical & Storage Services           | <b>Self Storage</b>                 | Self Storage Facilities                                     |
| 4  | Transport, Logistics & Utilities | 4.2   | Logistical & Storage Services           | <b>Cold Storage</b>                 | Refrigerated Logistics Facilities                           |
| 4  | Transport, Logistics & Utilities | 4.3.1 | Electricity, Gas & Thermal Distribution | <b>Energy Infrastructure</b>        | Substations, Grid Infrastructure                            |
| 4  | Transport, Logistics & Utilities | 4.3.2 | Water & Sewage Infrastructure           | <b>Water Infrastructure</b>         | Reservoirs, Treatment Plants                                |
| 4  | Transport, Logistics & Utilities | 4.3.3 | Waste Treatment                         | <b>Environmental Infrastructure</b> | Recycling Facilities, Waste Treatment Plants                |
| 5  | Residential Use                  | 5.1   | Permanent Residential Use               | <b>Residential</b>                  | Houses, Apartments, Residential Blocks                      |
| 5  | Residential Use                  | 5.1   | Permanent Residential Use               | <b>Multifamily</b>                  | Multifamily Housing                                         |
| 5  | Residential Use                  | 5.1   | Permanent Residential Use               | <b>Build to Rent (BTR)</b>          | Institutional Rental Housing                                |
| 5  | Residential Use                  | 5.1   | Permanent Residential Use               | <b>Luxury Residential</b>           | Prime Residential, Super Prime Residential                  |
| 5  | Residential Use                  | 5.2   | Residential with Compatible Uses        | <b>Mixed Use Living</b>             | Residential Led Mixed Use                                   |
| 5  | Residential Use                  | 5.3   | Other Residential Use                   | <b>PBSA</b>                         | Purpose-Built Student Accommodation                         |
| 5  | Residential Use                  | 5.3   | Other Residential Use                   | <b>Co-Living</b>                    | Shared Living Assets                                        |
| 5  | Residential Use                  | 5.3   | Other Residential Use                   | <b>Senior Living</b>                | Retirement Villages, Independent Living                     |
| 5  | Residential Use                  | 5.3   | Other Residential Use                   | <b>Serviced Living</b>              | Branded Residences, Serviced Apartments                     |
| 6  | Other Uses                       | 6.1   | Transitional Areas                      | <b>Development Land</b>             | Construction Sites, Regeneration Projects                   |
| 6  | Other Uses                       | 6.2   | Abandoned Areas                         | <b>Brownfield Regeneration</b>      | Redevelopment Opportunities                                 |
| 6  | Other Uses                       | 6.3   | Natural Areas                           | <b>Environmental Assets</b>         | Conservation Land, Biodiversity Assets                      |
| 6  | Other Uses                       | 6.4   | Areas Where Any Use Allowed             | <b>Strategic Land</b>               | Long-Term Development Land                                  |
| 6  | Other Uses                       | 6.5   | Areas Without Specified Planned Use     | <b>Unallocated Land</b>             | Future Planning Allocations                                 |
| 6  | Other Uses                       | 6.6   | Not Known Use                           | <b>Unknown</b>                      | Unclassified Assets                                         |